Code: 20A04504a **R20**

B.Tech III Year I Semester (R20) Regular & Supplementary Examinations November/December 2024

COMPUTER ARCHITECTURE & ORGANIZATION

(Electronics & Communication Engineering)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1 Answer the following: (10 X 02 = 20 Marks)

	(a)	How do you calculate the performance trade-offs when designing a processor?	2M
	(b)	Compare different memory types used in a computer system.	2M
	(c)	How would you implement a simple instruction fetch cycle?	2M
	(d)	Compare different types of addressing modes.	2M
	(e)	What is the range of number represented by sign?	2M
	(f)	What is the normalization in floating point number?	2M
	(g)	Illustrate how synchronous data transfer occurs between a CPU and an I/O device.	2M
	(h)	Analyze the advantages and disadvantages of vectored interrupts over regular interrupts.	2M
	(i)	How would you design a simple multi-core processor system?	2M
	(j)	Analyze the trade-offs between using shared memory and distributed memory in parallel computing systems.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2		Draw the block diagram and explain the functions of various components of digital computer.	10M
		OR	
3		Draw the control unit block diagram and explain the functionality of components in it.	10M
4		Analyze the impact of different addressing modes on instruction execution time.	10M
		OR	
5		Evaluate the pros and cons of using a micro programmed control unit.	10M
6		Distinguish between fixed point representation and floating point representation with an example.	10M
		OR	
7	(a)	What are the different data types and its representation.	5M
	(b)	Obtain the 1's complement & 2's complement for the numbers (i) –36 (8bit), (ii) 25.	5M
8		Compare different I/O data transfer techniques: Programmed I/O, Interrupt-driven I/O, and DMA.	10M
		OR	
9		Evaluate the effectiveness of synchronous vs. asynchronous data transfer in real-time systems.	10M
10		How would you optimize the performance of a pipelined processor in a modern computing environment?	10M
		OR	
11		Compare the performance and efficiency of using vector processors and scalar processors for high-performance computing tasks.	10M

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R20

B.Tech III Year I Semester (R20) Regular & Supplementary Examinations January 2024

COMPUTER ARCHITECTURE & ORGANIZATION

(Electronics & Communication Engineering)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1		Answer the following: (10 X 02 = 20 Marks)	
	(a)	Write about register transfer notations.	2M
	(b)	For what reason devices generate interrupts?	2M
	(c)	List the four basic functions of the CPU.	2M
	(d)	Write the address sequencing capabilities required in a control memory.	2M
	(e)	Explain the conversion of octal number to hexadecimal number with an example.	2M
	(f)	Design an Adder to add two 4-bit numbers.	2M
	(g)	Explain significance of Memory hierarchy.	2M
	(h)	Discuss about possible modes of data transfer.	2M
	(i)	What is Parallel processing?	2M
	(j)	Describe the need for inter processor communication.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2	Briefly explain the input-output instructions.	10M
	OR	
3	Draw and explain the working of a bus system for four registers.	10M
4	Briefly explain logical and bit manipulation, shift instructions.	10M
	OR	
5	Explain the working of microprogram sequencer with a neat diagram.	10M
6	Show the step by step multiplication process using booth algorithm when the following binary numbers are multiplied (+15)*(-13). Assume 5-bit registers that hold signed numbers and draw the flow chart for the corresponding example.	10M
	OR	
7	Draw and explain the addition and subtraction of floating-point numbers.	10M
8	Construct an Associative memory page table with number of words equal to the number of blocks in the main memory.	10M
	OR	
9	Discuss about Set Associative mapping.	10M
10	Explain instruction pipeline with neat timing diagram.	10M
	OR	
11	Explain briefly about the characteristics of multiprocessors.	10M

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B.Tech III Year I Semester (R20) Supplementary Examinations August 2023

COMPUTER ARCHITECTURE & ORGANIZATION

(Electronics & Communication Engineering)

Time: 3 hours Max. Marks: 70

PART – A

(Compulsory Question)

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1 Answer the following: (10 X 02 = 20 Marks)

(a)	What is computer Architecture?	2M
(b)	List the phases of instruction cycle.	2M
(c)	List the address sequencing capabilities required in a control memory.	2M
(d)	What are the fields in Instruction format?	2M
(e)	Find (1001101 - 10101001) using 1's complement?	2M
(f)	Multiply 10111 by 10011 using successive shift and add operations process.	2M
(g)	Write any two differences that exist between Central computer and Peripherals.	2M
(h)	What is Bootstrap Loader?	2M
(i)	What is the Flynn's classification of computers?	2M
(j)	Write a short note on synchronous bus.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2		Distinguish between circular shift and arithmetic shift with proper example.	10M
		OR	
3		Explain memory reference instructions with an examples.	10M
4		Explain the design of micro programmed control unit in detail.	10M
		OR	
5	(a)	Explain three address instruction formats.	5M
	(b)	Explain Immediate and Register Indirect Addressing modes.	5M
6		Convert the following binary number into decimal & octal number: (i) $(00010.110)_2$ (ii) $(000.10110)_2$.	10M
		OR	
7		Discuss about Booth's multiplication algorithm.	10M
8		Distinguish between Isolated versus Memory Mapped I/O.	10M
		OR	
9		Explain different types of mapping functions in cache memory.	10M
10		Distinguish the characteristics of RISC and CISC.	10M
		OR	
11		What is multiprocessor system? Explain the advantages of multi processors over uniprocessors.	10M

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